8.1.1 Basic concepts of sampling

Population

The group of individuals considered under study is called as population. The word population here refers not only to people but to all items that have been chosen for the study. Thus in statistics, population can be number of bikes manufactured in a day or week or month, number of cars manufactured in a day or week or month, number of fans, TVs, chalk pieces, people, students, girls, boys, any manufacturing products, etc…

Finite and infinite population:

When the number of observations/ individuals/products is countable in a group, then it is a finite population. Example: weights of students of class XII in a school.

When the number of observations/ individuals/products is uncountable in a group, then it is an infinite population. Example: number of grains in a sack, number of germs in the body of a sick patient.

Sample and sample size

A selection of a group of individuals from a population in such a way that it represents the population is called as sample and the number of individuals included in a sample is called the sample size.

Parameter and statistic

Parameter: The statistical constants of the population like mean (μ), variance(σ^2) are referred as population parameters.

Statistic : Any statistical measure computed from sample is known as statistic.

Note

In practice, the parameter values are not known and their estimates based on the sample values are generally used.

Types of sampling

The technique or method of selecting a sample is of fundamental importance in the theory of sampling and usually depends upon the nature of the data and the type of enquiry. The procedures of selecting a sample may be broadly classified as

1. Non-Random sampling or Non-probability sampling.
2. Random Sampling or Probability sampling.

Note

Here we confine ourselves to Random sampling or Probability sampling.

Random sampling or Probability sampling

Random sampling refers to selection of samples from the population in a random manner. A random sample is one where each and every item in the population has an equal chance of being selected.

" Every member of a parent population has had equal chances of being included".- Dr. Yates

"A random sample is a sample selected in such a way that every item in the population has an equal chance of being included".-Harper

The following are different types of probability sampling:

(i) Simple random sampling
(ii) Stratified random sampling
(iii) Systematic sampling

(i) Simple random sampling

In this technique the samples are selected in such a way that each and every unit in the population has an equal and independent chance of being selected as a sample. Simple random sampling may be done, with or without replacement of the samples selected. In a simple random sampling with replacement there is

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